1 Methods

1.1 Fine-scale H&E–MALDI-MSI alignment

We perform fine-scale alignment of MALDI-MSI measurements to H&E histology after an initial fiducial-based registration. Let

$$\Omega_{\mathrm{MSI}} = \{u_1, \dots, u_n\} \subset \mathbb{R}^2$$

denote the MSI measurement positions, and let

$$X \in \mathbb{R}^{n \times p}$$

be the corresponding preprocessed MSI feature matrix. We assume that an initial transformation

$$T_0 : \Omega_{\mathrm{MSI}} \to \mathbb{R}^2$$

maps MSI coordinates into the H&E coordinate system with an accuracy of a few MSI pixels. Fine alignment aims to estimate the remaining local deformation between both modalities.

As an anatomical reference signal, we extract a binary mask

$$S_{\mathrm{HE}}(z) \in \{0, 1\}$$

from the H&E image. The mask is chosen such that it represents a spatially informative tissue structure expected to have corresponding molecular signatures in the MSI data. In liver tissue, we use a sinusoid mask derived from H&E colour information. Since the native H&E resolution is substantially higher than the MSI resolution, the binary mask is converted into a continuous mask field

$$f(z)$$

in H&E coordinates by locally normalised Wendland kernel smoothing. Details on kernel construction, boundary correction and interpolation are given in Supplementary Section 2.1.

For any transformation T, the H&E-derived signal corresponding to an MSI pixel is obtained by evaluating the smooth mask at the transformed MSI coordinate,

$$s_T(i) = f(T(u_i)).$$

The resulting vector

$$s_T = (s_T(1), \dots, s_T(n))^\top$$

defines the H&E reference signal directly on the MSI measurement positions.

We next learn an MSI-derived representation of this anatomical signal. Using the initial transform T_0, we determine a linear MSI score

$$q = X\hat{w}^{(0)}$$

by solving

$$\hat{w}^{(0)} = \arg\max_{\|w\|_2 = 1} \rho(Xw, s_{T_0}),$$

where ρ denotes Pearson correlation. The optimisation is performed using a ridge-regularised regression formulation, and the ridge parameter is selected by spatial block cross-validation as described in Supplementary Section 2.2. The learned MSI score q is kept fixed during subsequent geometric optimisation to avoid simultaneous adaptation of image registration and MSI feature selection.

Residual geometric correction is performed in two stages. We first fit a globally smooth displacement field using thin-plate splines,

$$T_1(u_i) = T_0(u_i) + \Delta^{\mathrm{TPS}}(u_i).$$

This step captures broad residual deformation and provides a stable first-pass estimate of the displacement field. However, globally smooth transformations cannot accurately represent discontinuous tissue motion caused by tears or ruptures. Therefore, the TPS model is not used as the final deformation model.

Instead, the TPS displacement field is used to identify locations where the assumption of smooth deformation is violated. Tissue discontinuities are expected to appear as narrow regions in which neighbouring MSI pixels require substantially different displacement vectors. We therefore derive local rupture scores from spatial changes in the TPS displacement field and convert them into weights describing the confidence that neighbouring pixels belong to the same continuously deforming tissue region. The calculation of rupture scores and regularisation weights is described in Supplementary Section 2.3.

The final alignment is obtained by fitting a dense residual displacement field relative to the TPS-corrected transform,

$$\hat{T}(u_i) = T_1(u_i) + \hat{\Delta}_i.$$

The residual field is regularised using a graph-Laplacian penalty on neighbouring MSI pixels. This penalty enforces locally smooth deformations within intact tissue regions but is weakened near candidate ruptures, allowing discontinuous motion between separated tissue fragments. In addition, pixels located directly within likely rupture regions receive reduced weight in the alignment objective. Details of the weighted optimisation problem are given in Supplementary Section 2.3.

The discontinuity-adaptive refinement is performed once. Specifically, the smooth TPS fit is used to estimate rupture probabilities, these probabilities are fixed, and a single graph-Laplacian regularised displacement fit is performed. We do not iteratively update rupture locations during optimisation, since such feedback may allow the model to explain local noise by introducing artificial discontinuities.

After convergence, the MSI feature weights are re-estimated using the final transformation,

$$\hat{w}^{(\mathrm{final})} = \arg\max_{\|w\|_2=1} \rho(Xw, s_{\hat{T}}).$$

This final fit is used only as a diagnostic. We compare the initial and final MSI scores,

$$q^{(0)} = X\hat{w}^{(0)}, \qquad q^{(\mathrm{final})} = X\hat{w}^{(\mathrm{final})},$$

by their absolute Pearson correlation,

$$\rho_{\mathrm{score}} = \left|\rho(q^{(0)}, q^{(\mathrm{final})})\right|.$$

This score-level comparison is preferred over direct comparison of the weight vectors, because correlated MSI features may yield different coefficient vectors representing nearly identical spatial patterns.

Registration quality is assessed by the improvement in H&E–MSI agreement, the magnitude and spatial structure of the estimated displacement field, the inferred rupture scores, and the agreement between the initial and final MSI scores.

2 Supplementary Methods

2.1 Wendland smoothing and interpolation of the H&E mask

Let

$$S_{\mathrm{HE}}(z) \in \{0,1\}, \qquad z \in \Omega_{\mathrm{HE}},$$

denote the binary H&E-derived mask on the H&E pixel grid. To obtain a continuous objective function for image registration, the binary mask is converted into a smooth field using convolution with a compactly supported Wendland kernel.

For a support radius ρ, we define

$$K_\rho(x) = \frac{1}{C_\rho}\left(1 - \frac{\|x\|_2^2}{\rho^2}\right)_+^4 \left(4\frac{\|x\|_2^2}{\rho^2} + 1\right),$$

where $(a)_+ = \max(a, 0)$ and C_ρ normalises the discrete kernel such that

$$\sum_x K_\rho(x) = 1.$$

The kernel is evaluated only for offsets satisfying

$$\|x\|_2 < \rho.$$

The support radius is chosen from the sampling density of the transformed MSI grid. Let

$$p_i = T_0(u_i)$$

be the MSI pixel centres mapped into H&E coordinates. We estimate the local MSI spacing by

$$d_{\mathrm{MSI}} = \mathrm{median}_{(i,j)\in\mathcal{N}} \|p_i - p_j\|_2,$$

where $\mathcal{N}$ contains neighbouring MSI pixels on the original MSI grid. Assuming that T_0 is locally approximately rigid, the covering radius of the transformed MSI grid is

$$d_{\mathrm{MSI}}/\sqrt{2}.$$

We therefore set

$$\rho = (1+\eta)\frac{d_{\mathrm{MSI}}}{\sqrt{2}},$$

with a small safety factor $\eta = 0.1$. This ensures that every H&E pixel inside the mapped MSI footprint contributes to at least one transformed MSI measurement location.

Because tissue masks may have irregular boundaries, ordinary zero-padded convolution would incorrectly treat regions outside the valid H&E domain as background. We therefore use locally normalised convolution. Let

$$M(z) \in \{0,1\}$$

be the valid-pixel mask of the H&E image. We compute

$$N(z) = (S_{\mathrm{HE}} * K_\rho)(z), \qquad D(z) = (M * K_\rho)(z),$$

where $*$ denotes discrete convolution after zero extension outside the valid image domain. The final smoothed mask is

$$f(z) = \frac{N(z)}{D(z)}$$

for all positions with $D(z) > 0$.

For arbitrary transformed MSI coordinates $T(u_i)$, which generally do not coincide with H&E pixel centres, the value

$$s_T(i) = f(T(u_i))$$

is obtained by bilinear interpolation of the discrete field f.

2.2 MSI metafeature learning and ridge selection

For a fixed transformation T, the goal is to determine an MSI feature combination

$$q = Xw$$

that maximally agrees with the H&E-derived soft mask s_T. The canonical correlation problem

$$\hat{w} = \arg\max_{\|w\|_2=1} \rho(Xw, s_T)$$

is solved through its equivalent ridge-regularised regression formulation.

All MSI features are first centred and scaled to unit variance. We then solve

$$\hat{\beta}_\lambda = \arg\min_\beta \left\{ \|s_T - X\beta\|_2^2 + \lambda\|\beta\|_2^2 \right\}.$$

The corresponding canonical direction is

$$\hat{w}_\lambda = \frac{\hat{\beta}_\lambda}{\|\hat{\beta}_\lambda\|_2}.$$

The ridge parameter λ is selected by spatial block cross-validation. The MSI measurement area is divided into approximately regular spatial blocks. Small boundary blocks containing too few measured pixels are merged with neighbouring blocks. The resulting blocks are randomly assigned to five folds while approximately balancing the number of MSI pixels per fold.

For each candidate value of λ, the model is fitted on four folds. The resulting MSI score is evaluated on the held-out fold by

$$\rho_{\text{val}}(\lambda) = \rho(X_{\text{val}}\hat{w}_\lambda, s_{T,\text{val}}).$$

The selected ridge parameter maximises the average validation correlation across folds.

2.3 Discontinuity-adaptive displacement optimisation

Starting from the initial transform T_0, we first estimate a smooth thin-plate spline (TPS) correction

$$T_1(u_i) = T_0(u_i) + \Delta^{\text{TPS}}(u_i).$$

The TPS displacement is parameterised by a sparse set of control points on the MSI grid and is estimated by solving

$$\hat{\Delta}^{\text{TPS}} = \arg\max_{\Delta^{\text{TPS}}} \left[\rho(q, s_{T_1}) - \lambda_{\text{TPS}} E_{\text{bend}}(\Delta^{\text{TPS}}) \right],$$

subject to loose displacement bounds. Here E_{bend} denotes the classical TPS bending energy. This first step provides a globally smooth approximation of the residual deformation. It is not used as the final deformation model, because global smoothness assumptions are violated in the presence of tissue ruptures.

The TPS displacement field is instead used to identify locations where the assumption of smooth deformation is locally violated. Let

$$\Delta_i^{\mathrm{TPS}} = \Delta^{\mathrm{TPS}}(u_i).$$

For neighbouring MSI pixels $i \sim j$, we calculate

$$g_{ij} = \frac{\|\Delta_i^{\mathrm{TPS}} - \Delta_j^{\mathrm{TPS}}\|_2}{\|u_i - u_j\|_2}.$$

This quantity measures local changes in the displacement field. Tissue ruptures are expected to appear as spatially coherent ridges of high values. A vertex-level score is obtained by

$$G_i = \max_{j:j\sim i} g_{ij}.$$

The scores are robustly standardised,

$$Z_i = \frac{G_i - \mathrm{median}(G)}{1.4826\,\mathrm{median}_k\,|G_k - \mathrm{median}(G)|}.$$

The resulting Z_i values are interpreted as continuous rupture evidence scores rather than as binary rupture calls.

The rupture scores are converted into local non-rupture weights

$$p_i = \frac{1}{1 + \exp(Z_i - z_0)},$$

with default value

$$z_0 = 4.$$

Thus, pixels with small rupture evidence have $p_i \approx 1$, whereas candidate rupture pixels have reduced values.

The graph edge weights used for the final smoothness penalty are defined as

$$a_{ij} = a_{\min} + (1 - a_{\min}) \min(p_i, p_j),$$

with

$$a_{\min} = 0.05.$$

The lower bound prevents individual pixels or small regions from becoming completely disconnected from the displacement graph, while still allowing substantially reduced coupling near rupture candidates.

Pixels located inside candidate rupture regions can optionally be downweighted in the alignment objective. We define

$$r_i = p_i^{\beta},$$

where $\beta \geq 0$ controls the strength of this downweighting. The default choice

$$\beta = 0$$

keeps all pixels equally weighted in the correlation objective and uses the rupture information only to adapt the deformation regularisation. Larger values of β progressively reduce the contribution of pixels located in likely rupture regions.

The final residual transformation is

$$T_\Delta(u_i) = T_1(u_i) + \Delta_i,$$

where Δ_i is a dense displacement field defined on the MSI grid. It is estimated by

$$\hat{\Delta} = \arg\max_{\Delta} \left[\rho_r(q, s_\Delta) - \lambda \sum_{i\sim j} a_{ij} \|\Delta_i - \Delta_j\|_2^2 \right],$$

subject to loose displacement bounds

$$\|\Delta_i\|_\infty \le d_{\max}.$$

The soft mask values are

$$s_\Delta(i) = f(T_1(u_i) + \Delta_i),$$

where f is evaluated by bilinear interpolation.

The weighted Pearson correlation is defined as

$$\rho_r(q, s_\Delta) = \frac{\sum_i r_i(q_i - \bar{q}_r)(s_\Delta(i) - \bar{s}_r)}{\sqrt{\sum_i r_i(q_i - \bar{q}_r)^2}\sqrt{\sum_i r_i(s_\Delta(i) - \bar{s}_r)^2}},$$

with

$$\bar{q}_r = \frac{\sum_i r_i q_i}{\sum_i r_i}, \qquad \bar{s}_r = \frac{\sum_i r_i s_\Delta(i)}{\sum_i r_i}.$$

For the default choice $\beta = 0$, this reduces to the ordinary Pearson correlation.

No affine component, absolute displacement penalty, or gauge constraint is introduced. Constant displacement fields are therefore unpenalised and can correct remaining global shifts. The regularisation acts only on spatial variation of the displacement field.

The rupture weights are estimated once from the TPS displacement field and then kept fixed during the graph-Laplacian refinement. The procedure is deliberately not iterated to avoid feedback in which the deformation model creates artificial discontinuities that are subsequently reinforced.